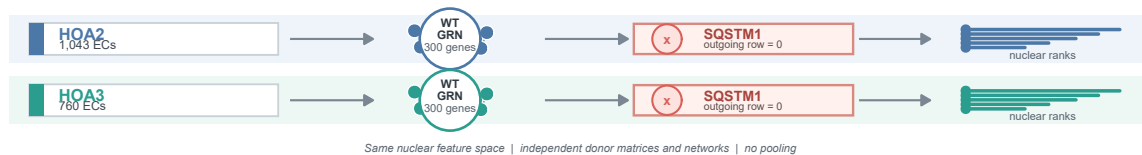


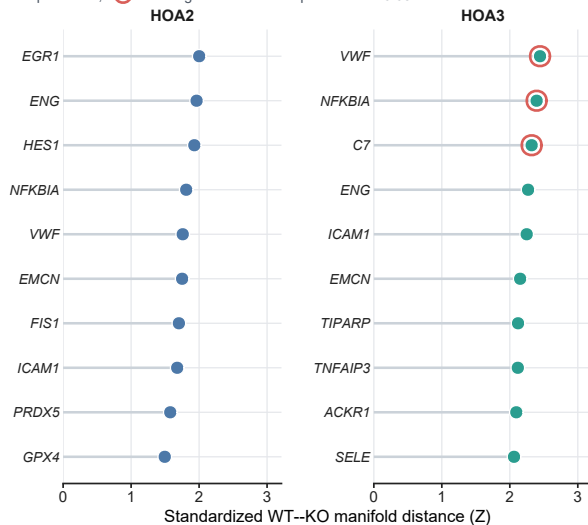
A Independent donor-resolved virtual knockout

Matched feature space; separate networks, perturbations and readouts



B Predicted nuclear differential regulation

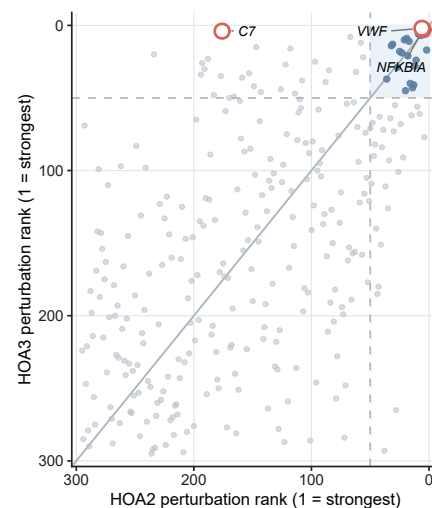
Top 10 downstream ranks per donor; ○ red rings denote donor-specific FDR<0.05



C Cross-donor rank agreement

Spearman rho=0.555; shared top-50=22; replicated FDR hits=0

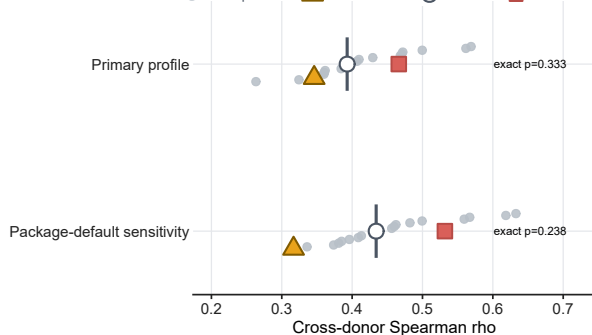
○ Red rings: HOA3-only hits



D

Receptor-context specificity calibration

● Comparators ▲ CALCOCO2 ○ Median ■ SQSTM1



E

Matched-null pathway rank recovery

□ Red outline: empirical FDR<0.05 in one donor only; no pathway replicated

